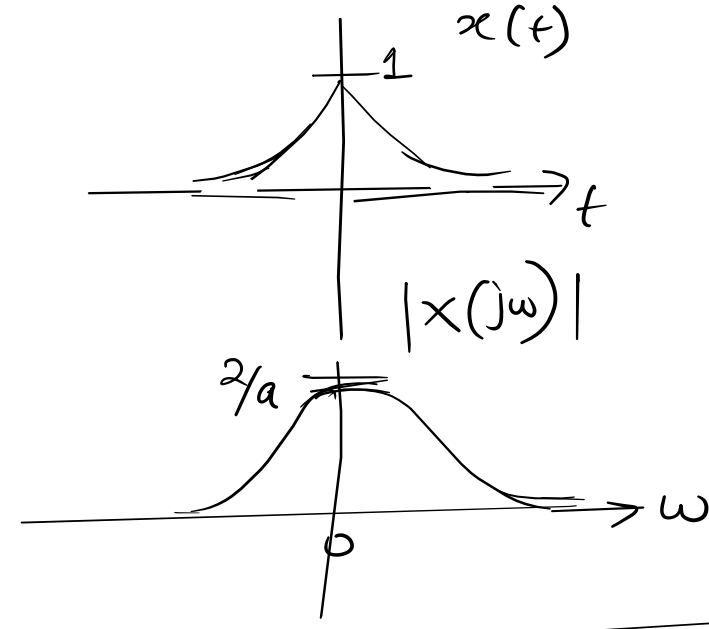
Fourier Transform

15/10/2025

Ex $x(t) = e^{-a|t|} \quad a > 0$

$$X(j\omega) = \frac{2a}{a^2+\omega^2}$$



if $a \longrightarrow 0$

$x(t) \longrightarrow 1 \quad \forall \quad t$

$1 \xrightarrow[\text{I.F.T}]{\text{F.T.}} 2\pi\delta(\omega)$

$\delta(t) \rightleftharpoons 1$

T. P. T $\quad X(j\omega) \longrightarrow 2\pi\delta(\omega)$

$$\int_{-\infty}^{\infty} X(j\omega)\,d\omega = \int_{-\infty}^{\infty} \frac{2a\,d\omega}{a^2+\omega^2}$$

let $\omega = a\tan\theta$

$d\omega = a\sec^2\theta\, d\theta$

$$= \int_{-\pi/2}^{\pi/2} \frac{2a^2\sec^2\theta\, d\theta}{a^2(1+\tan^2\theta)} = \int_{-\pi/2}^{\pi/2} 2\,d\theta = 2\pi$$

$$\lim_{a\to 0} X(j\omega) = \lim_{a\to 0} \frac{2a}{a^2+\omega^2} \longrightarrow 0 \qquad \omega \neq 0$$

$$\Longrightarrow \infty \quad \text{if} \quad \omega = 0$$

Area under $x(j\omega) = 2\pi$

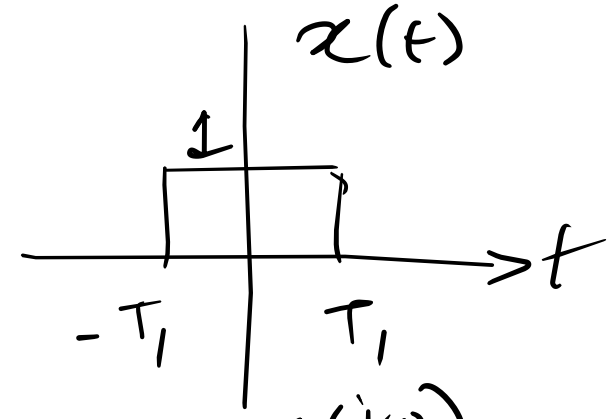
$$\lim_{a\to 0} x(j\omega) = 2\pi\,\delta(\omega)$$

$$= \begin{cases} 0 & \omega \neq 0 \\ \infty & \end{cases}$$

$$\& \int_{-\infty}^{\infty} X(j\omega)\,d\omega = 2\pi$$

$a \to 0$, $x(t) \to 1$ & $X(j\omega) \longrightarrow 2\pi\delta(\omega)$

Ex $\quad x(t) = \begin{cases} 1 & -T_1 \le t \le T_1 \\ 0 & \text{otherwise} \end{cases}$



$$X(j\omega) = \frac{2\sin\omega T_1}{\omega}$$

$T_1 \longrightarrow \infty$

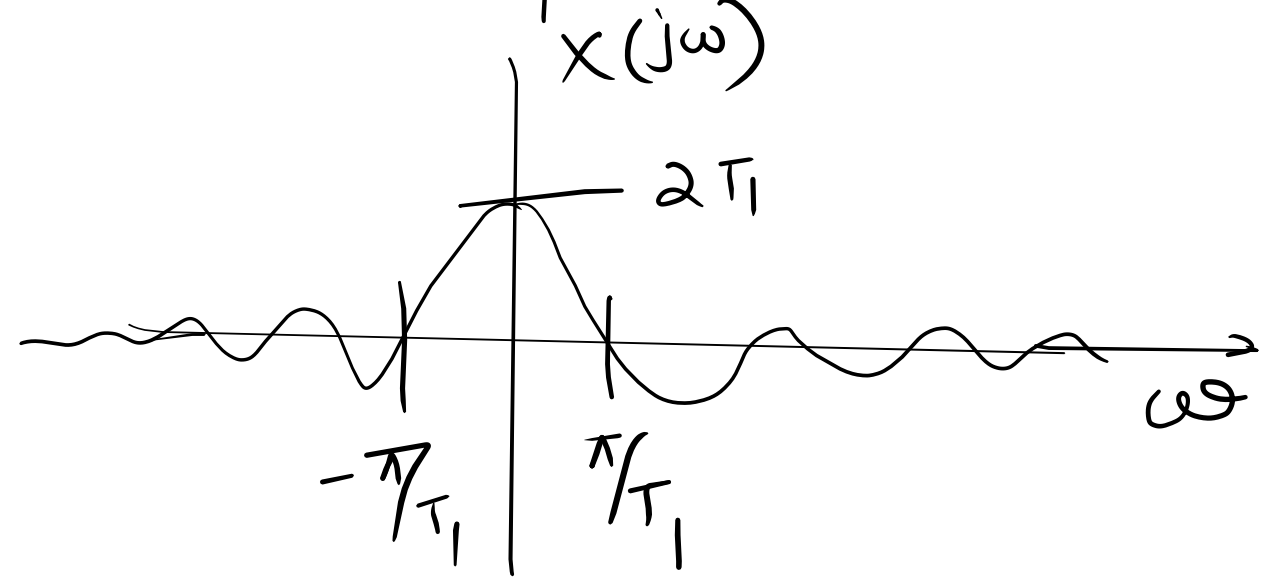
$x(t) \longrightarrow 1 \quad \forall\, t$

$X(j\omega) \longrightarrow 0 \qquad \omega \neq 0$

$\longrightarrow 2T_1 \qquad \omega = 0$

$\longrightarrow$ undefined

$\longrightarrow \infty$



$$\int_{-\infty}^{\infty} X(j\omega)\, d\omega = \text{Constant} = 2\pi$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

$$2\pi\, x(t) = \int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

$t = 0$

$$2\pi\, x(0) = \boxed{2\pi = \int_{-\infty}^{\infty} X(j\omega)\, d\omega}$$

as $T_1 \to \infty$

$x(t) \longrightarrow 1 \quad \forall\, t$

$X(j\omega) \longrightarrow 2\pi\,\delta(\omega)$

Ex

$$x(t) = \begin{cases} e^{-at} & t > 0 \\ 0 & t = 0 \\ -e^{at} & t < 0 \end{cases}$$

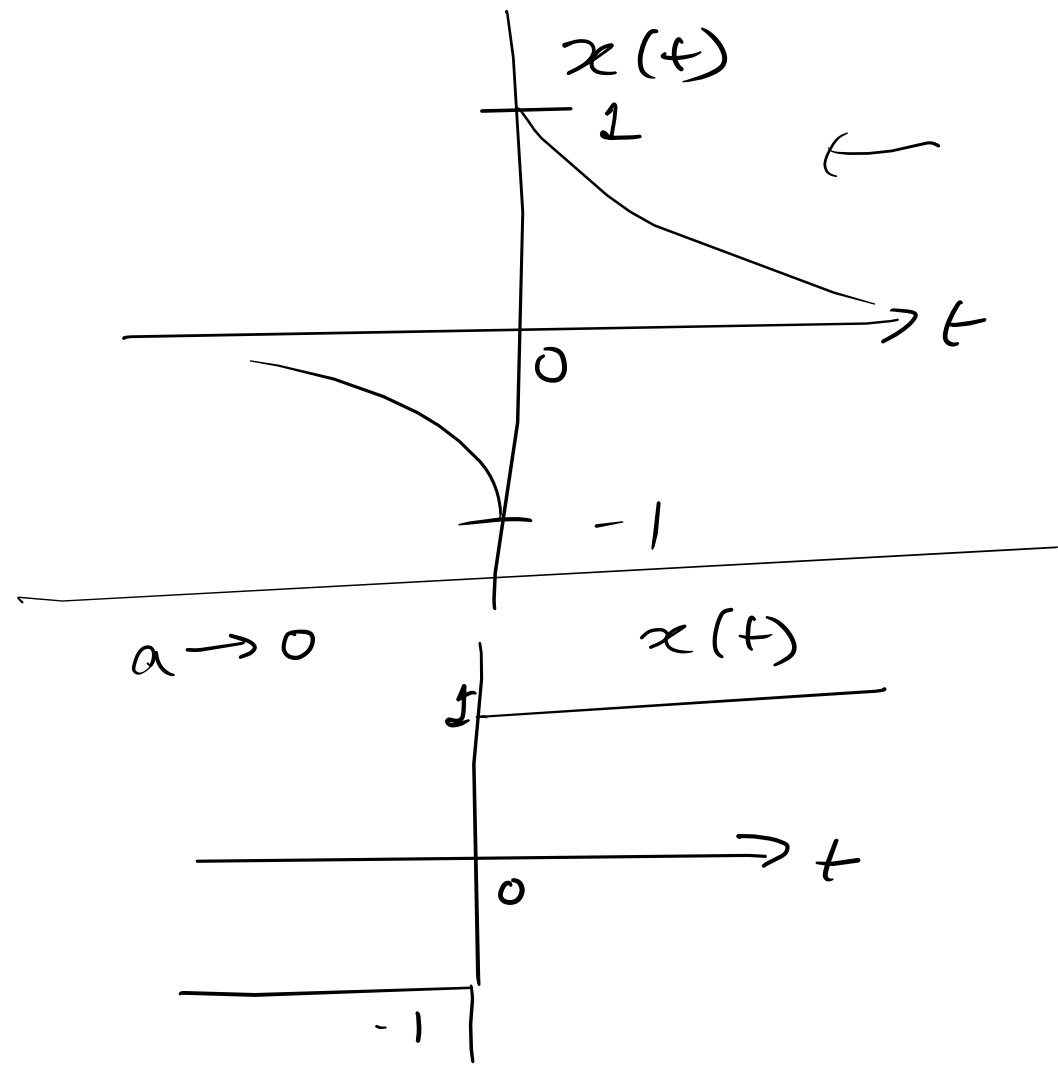
as $a \longrightarrow 0$

$x(t) \longrightarrow Sgn(t)$

$$= \begin{cases} 1 & t > 0 \\ 0 & t = 0 \\ -1 & t < 0 \end{cases}$$

$$u(t) = \frac{1}{2}\left(Sgn(t) + 1\right)$$

$$= \begin{cases} 1 & t > 0 \\ 1/2 & t = 0 \\ 0 & t < 0 \end{cases}$$

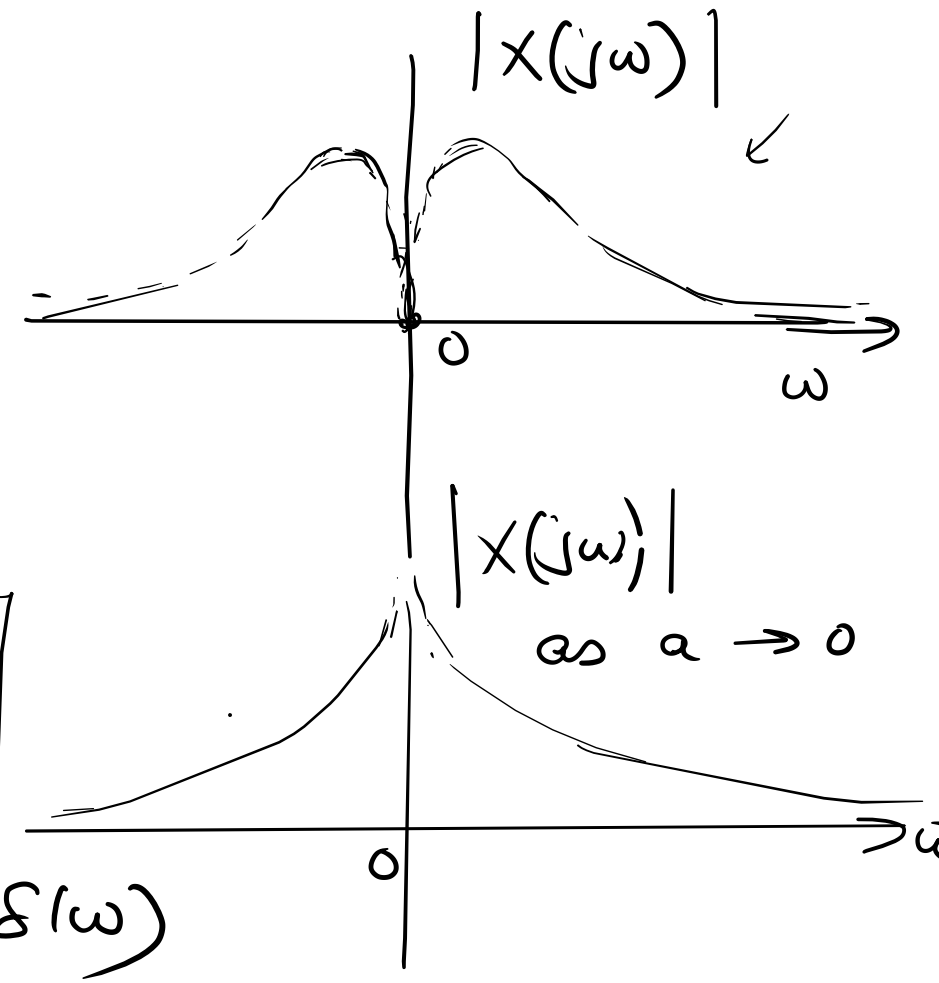


$$X(j\omega) = \int_{-\infty}^{0} -e^{at} e^{-j\omega t}\, dt + \int_{0}^{\infty} e^{-at} \cdot e^{-j\omega t}\, dt$$

$$X(j\omega)\Big|_{\omega=0} = \left.\frac{-2j\omega}{a^2+\omega^2}\right|_{\omega=0} = 0$$

$$a \longrightarrow 0 \qquad \lim_{a\to 0} X(j\omega) = \frac{-2j\omega}{\omega^2}$$

$$Sgn(t) \rightleftharpoons \boxed{\lim_{a\to 0} X(j\omega) = \frac{2}{j\omega}}$$



$$u(t) = \frac{1}{2}\left(Sgn(t)+1\right) \rightleftharpoons \frac{1}{j\omega} + \pi\delta(\omega)$$

$$\boxed{u(t) \rightleftharpoons \frac{1}{j\omega} + \pi\delta(\omega)}$$

Properties of F.T.

① linearity

$$x(t) \underset{IFT}{\overset{F.T.}{\rightleftharpoons}} X(j\omega)$$

$$y(t) \rightleftharpoons Y(j\omega)$$

$$z(t) = \alpha x(t) + \beta y(t) \rightleftharpoons \alpha X(j\omega) + \beta Y(j\omega)$$

$$Z(j\omega) = \int_{-\infty}^{\infty} z(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} \left(\alpha x(t) + \beta y(t)\right) e^{-j\omega t} dt$$

$$= \alpha \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt + \beta \int_{-\infty}^{\infty} y(t) e^{-j\omega t} dt$$

$$= \alpha X(j\omega) + \beta Y(j\omega)$$

$$=$$

(2) Time shifting $x(t) \rightleftharpoons X(j\omega)$

$$x(t-t_0) \rightleftharpoons e^{-j\omega t_0} X(j\omega)$$

$y(t) = x(t-t_0)$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$$Y(j\omega) = \int_{-\infty}^{\infty} y(t)\, e^{-j\omega t}\, dt = \int_{-\infty}^{\infty} x(t-t_0)\, e^{-j\omega t}\, dt$$

$t - t_0 = z$

$t = z + t_0$

$dt = dz$

$$= \int_{-\infty}^{\infty} x(z)\, e^{-j\omega(z+t_0)}\, dz$$

$$= e^{-j\omega t_0} \int_{-\infty}^{\infty} x(z)\, e^{-j\omega z}\, dz = e^{-j\omega t_0} X(j\omega)$$

$x(t)$ $\xrightarrow{\text{delayed}}$ $|X(j\omega)|$ remains the same.

Hence, FT is also called as a shift-invariant operator.

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(3) Shifting in the frequency domain

$$x(t) \rightleftharpoons X(j\omega)$$

$$e^{j\omega_0 t}\, x(t) = ? \rightleftharpoons X(j\omega - j\omega_0) = Y(j\omega)$$

given

Find actual value.

④ multiplication in time domain = convolution in the freq. domain

Convolution in time domain = multiplication in the freq. domain

$$x(t) \rightleftharpoons X(j\omega)$$

$$h(t) \rightleftharpoons H(j\omega)$$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \rightleftharpoons Y(j\omega) = X(j\omega)H(j\omega)$$

$$z(t) = x(t)y(t) \rightleftharpoons \frac{1}{2\pi}\{X(j\omega) * Y(j\omega)\}$$